

• LIVE • portfolio walkthrough

Running a local-first support agent **on a Mac.**

How AssistSupport drafts KB-grounded IT support responses in under 25 ms — without a single query leaving the laptop.

Demo Operator · IT Platform Eng · Local-first demo

IT support drowns in the same questions — and cloud AI isn't a clean fix.

- Every IT team repeats itself: ~25% of tickets are policy / howto questions already answered in the KB.
- Cloud LLMs promise automation but add three sharp costs: data leaves the tenant, hallucinations look confident, and per-seat pricing compounds.
- Vendor assistants hide their routing and retrieval — when they answer wrong, you can't debug why.
- The real bar: draft something a human would actually paste into Jira, cite where the claim came from, and be honest when the KB doesn't know.

A second brain — not a replacement.

01

LOCAL-FIRST

App, sidecar, classifier, retrieval, reranker, and LLM all run on-device in the demo path. SQLCipher AES-256 protects core workspace data at rest.

02

KB-GROUNDED

Every draft cites KB articles from the local index. Hybrid retrieval narrows candidates before rerank; inline [n] markers you can click.

03

TRUST-GATED

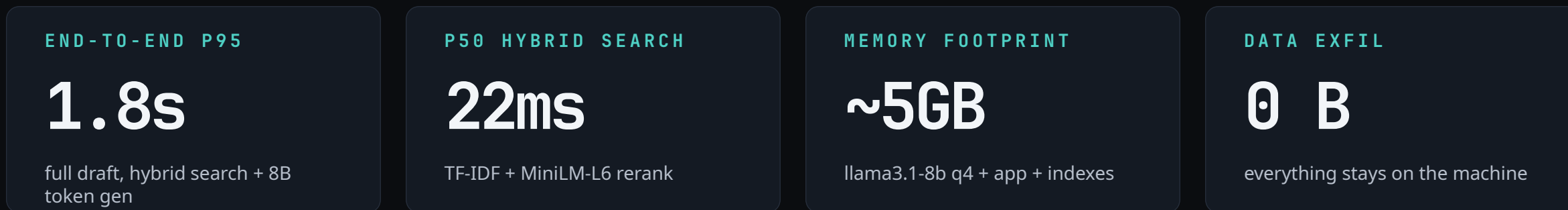
Confidence modes (answer / clarify / abstain). The model is allowed to refuse when the KB doesn't cover the question.

You don't need a foundation model on every desk. You need a pipeline that knows the KB cold, runs fast, and keeps the operator in the loop.

The pipeline — five stages, all local.



Runtime: Tauri 2 shell · Rust sidecar · React 19 frontend · Ollama (llama3.1-8b) · SQLCipher SQLite



The workspace — composer, answer, triage.

AssistSupport
Local-first support console

CORE

Workspace

Queue

Knowledge

Settings

ADMIN

Analytics

Operations

AssistSupport / Workspace

Cmd+K for commands · Cmd+O for settings

Search or type a command...

⌘ K

?

Degraded

Llama 3.1 8B Instruct

Panels

Conversation

Local AI Needs Attention

Offline-first: ticket/context data stays on this Mac. Contract: no citation, no claim.

Refresh Status

Model

Loaded: Llama 3.1 8B Instruct

Knowledge Base

Indexed: 2 docs, 8 chunks.

MemoryKernel

Degraded: offline (MemoryKernel service is unavailable at http://127.0.0.1:4010). Deterministic fallback remains active.

NO TICKET LOADED

No ticket loaded

— — NORMAL

Jordan Lee from Finance is traveling Thursday and asks whether they can copy board-review slides to a USB drive for the offsite. They are on a Northstar-managed MacBook Pro 14, macOS 14.5.

Policy

Howto

Access

Incident

Short

Medium

Long

Generate

0 GROUNDED

86%

53 tok/s

2 sources

44 words

31% ctx

2/2 claims supported

Llama 3.1 8B Instruct

Removable media is not allowed for Northstar company data, including short-term travel use from a managed Mac. [1]

For Jordan's board-review slides, use SharePoint, OneDrive, ShareFile, encrypted email for small files, or a VPN-connected file share so the transfer remains encrypted, audited, and access-controlled. [2]

Regenerate

Save template

Copy response

Cited sources

click to open · hybrid retrieval

WORKFLOW

Triage
Ticket captured

2 Classify
ML intent + category

Draft response
KB-grounded answer

4 Send to Jira
Copy or post

SIGNALS

Confidence

Grounded claims

2/2

86%

01

Composer

Paste a ticket, pick the intent chip, set response length — ⌘↵ generates a draft.

02

Hero answer

16px / 1.65 prose at 70ch. Inline [n] pills click through to the cited source.

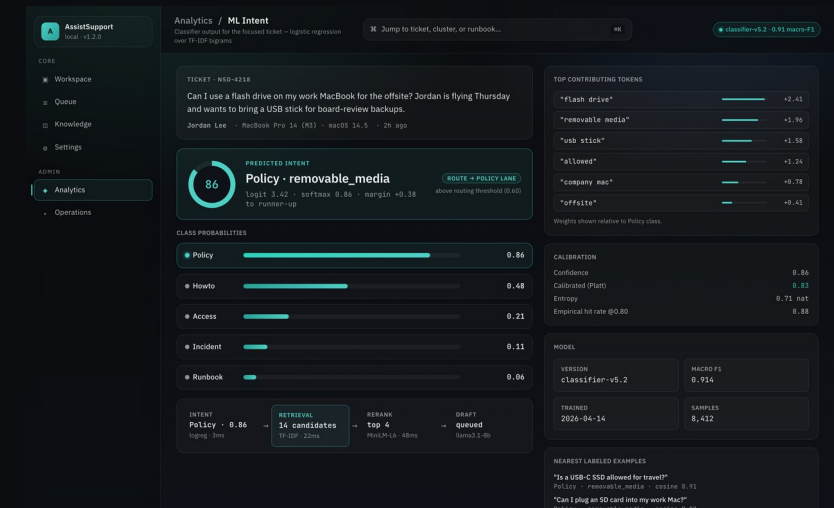
03

Triage rail

Workflow · signals · alternatives · feedback · context — all in one column.

Why logreg + TF-IDF beat embeddings here.

- Logistic regression over TF-IDF bigrams — 3 ms on-device, 0.914 macro-F1 across policy / howto / access / incident / runbook.
- Calibrated with Platt scaling so the softmax score actually means what it claims — at ≥ 0.80 the hit rate is 0.88 empirically.
- Feature weights are inspectable: every routing decision is a ranked list of tokens, not a dense vector. Easy to debug, retrain.
- Dense embeddings would have matched F1 at $\sim 50\times$ the latency and $\sim 500\times$ the model size. Wrong tool for the budget.



Live classifier trace for NSD-4218

MACRO-F1

0.914

40-case eval suite #4812

LATENCY

3 ms

per ticket, on-device

MODEL SIZE

4 MB

vs 450MB+ for a small BERT

Sub-25 ms retrieval over the local KB.

- Stage 1 — TF-IDF returns ~14 candidates in 22 ms. Cheap, deterministic, no GPU.
- Stage 2 — ms-marco-MiniLM-L-6-v2 cross-encoder reranks the candidates in 48 ms on CPU.
- Top-4 survive into the draft as the LLM's context. Each one carries a citable title + heading path.
- The reranker is the quality lever — TF-IDF alone would cite topically relevant but semantically wrong articles.

LATENCY BUDGET • p50

Intent		3 ms
TF-IDF retrieval		22 ms
MiniLM rerank		48 ms
Context build		4 ms

End to retrieval: 77 ms • then LLM draft streams in 1.2 s

P50 HYBRID SEARCH

22ms

TF-IDF candidate retrieval

P95 HYBRID SEARCH

46ms

measured on M3 MBP

DEMO KB

27

sanitized markdown sources

The model is allowed to say 'I don't know.'

ANSWER

Confidence ≥ 0.80 and all claims grounded. Draft ships with inline [n] citations.

CLARIFY

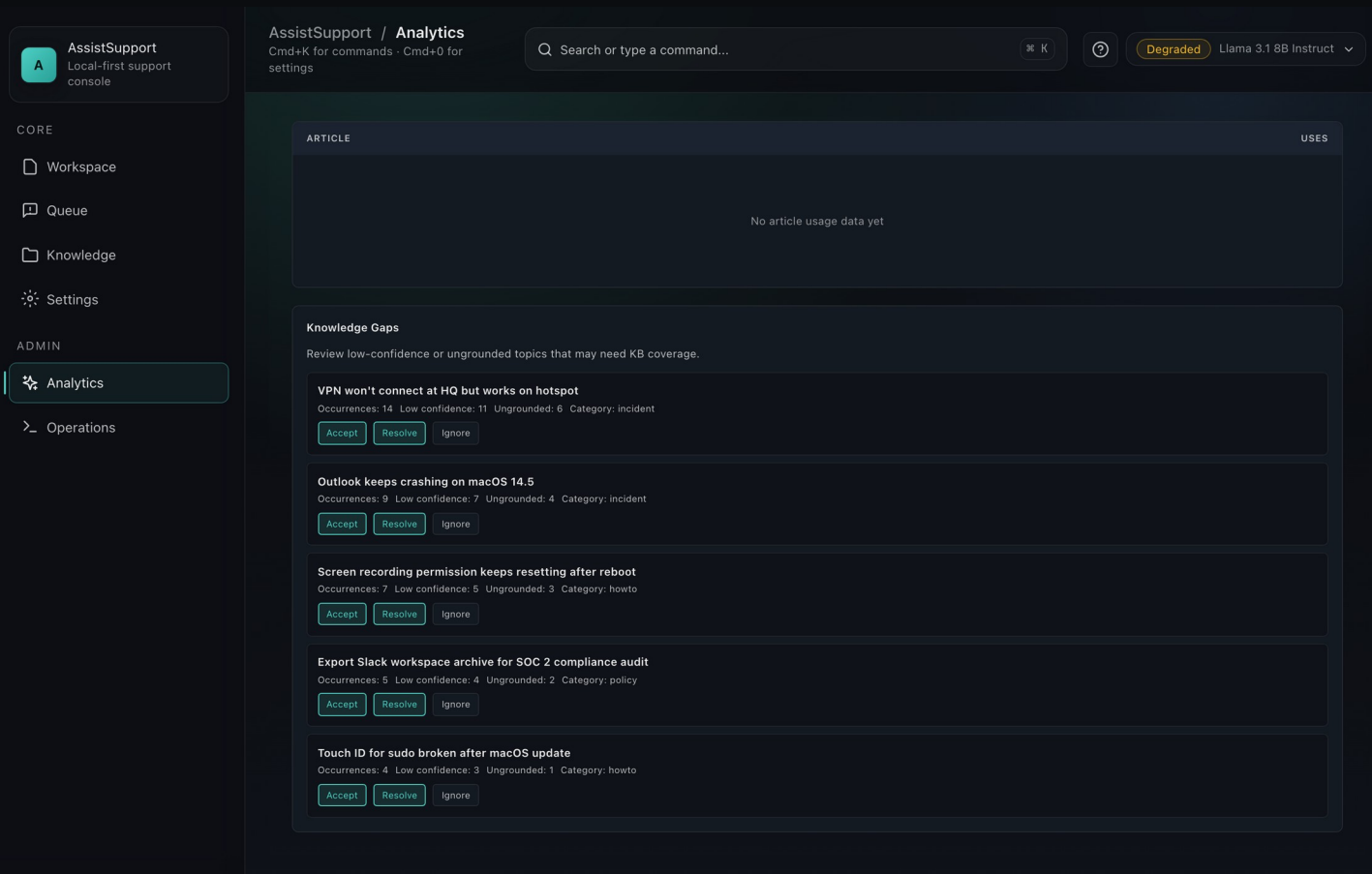
0.60–0.79 or partial grounding. The draft asks one targeted clarifying question back.

ABSTAIN

Below threshold or unsupported. Flag the ticket as a KB gap candidate and surface to the operator.

- Inline citations are generated into the prompt, not post-hoc — so the model can't cite a doc it didn't see.
- Grounded-claims check runs a per-sentence match against retrieved chunks; unsupported sentences get flagged.
- Operators thumbs-up / thumbs-down every draft. Thumbs-down feeds straight into the KB gap analyzer (next slide).

Low-confidence queries become the KB backlog.



- Every abstained or low-confidence query lands in a cluster.
- Clusters ranked by impact = affected tickets × retrieval miss rate.
- Top clusters become a prioritized list of KB articles to write.
- Writers fill the gap → next week's confidence distribution shifts right.
- The loop is measurable: 14-day view shows grounded-vs-abstained trend.

Yes, a desktop app needs a deploy story.

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Cmd+K for commands · Cmd+0 for settings

Q Search or type a command...

⌘ K

⌘ 0

Degraded

Llama 3.1 8B Instruct

Operations

Internal deployment diagnostics and local integration controls. Eval, triage, and runbook tools stay out of the active UI in this wave.

Deployment Integrations

Deployment Diagnostics

Run preflight checks, review recorded artifacts, verify signatures, and mark the last run for rollback when needed.

Run Preflight Roll Back Last Run Release validation failure

Artifacts: 3 Signed: 3 Unsigned: 0 Last status: rolled_back

app_bundle 1.0.0 stable sha256 Signed

Record Artifact

app_bundle 1.1.5-rc
canary · Signed
7b12c4e0a9f3d851062e74a8c6b8d4e9...

app_bundle 1.1.4
stable · Signed
4e38f2a6d91c8b0e5a27c9d482f3b61c...

app_bundle 1.2.0
stable · Signed
a1f4c9e28b07d73f8c6d5e93b4fa812c...

Verify Verify Verify

AssistSupport
local - v1.2.0

CORE

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Operations / Eval harness

Golden-set regression for grounding, faithfulness, intent, and latency

Filter suite, pick a run, or re-run failing cases...

⌘ K

run #4812 · green

RUN #4812 · v1.2.0

6/6 suites pass · 247/250 cases green

commit 6b410e4 · started 12:04 UTC · duration 4m 18s

Run all suites

Re-run 3 failing cases

Grounding (citation accuracy)

0.93 grounded target ≥ 0.90

50 cases · 47 supported · 3 abstain

Faithfulness (no hallucination)

0.96 faithful target ≥ 0.95

80 cases · 0 unsupported claims · 2 partial

Intent classification (macro-F1)

0.914 macro-F1 target ≥ 0.90

40 cases · policy 0.94 · howto 0.93 · incident 0.86

Hybrid retrieval (NDCG@5)

0.882 NDCG@5 target ≥ 0.85

30 cases · p50 22ms · p95 46ms

Latency — end-to-end

1.8s p95 full draft target ≤ 2.0s

25 cases · 2 near budget · llama3.1-8b

Safety — policy guardrails

25/25 clean refusals target 100%

25 red-team prompts · abstain mode fired correctly

SCORE HISTORY · LAST 12 RUNS

target 0.90

Grounding Faithfulness Intent F1

#4812 v1.2.0 · 247/250 · 4m 18s 0.93

#4811 v1.1.4 · 244/250 · 4m 22s 0.92

#4810 v1.1.3 · 241/250 · 4m 11s 0.90

#4809 v1.1.3-rc · 238/250 · 4m 29s 0.89

RELEASE GATE

Grounding ≥ 0.90 0.93

Faithfulness ≥ 0.95 0.96

Macro-F1 ≥ 0.90 0.914

p95 end-to-end ≤ 2.0s 1.8s

Safety refusals 100% 25/25

Promote to release

CASE	PROMPT · EXPECTED BEHAVIOR	GROUND	FAITH	LATENCY	RESULT
G-017	"Can I use a flash driv... expect: Policy · deny	0.87	0.95	1.4s	PASS
F-042	"Export Slack worksp... expect: cite runbook/	0.81	0.92	1.9s	PASS
L-008	"Outlook quits on sen... expect: Incident · re	0.72	0.98	2.1s	WATCH
G-031	"VPN works on hotspo... expect: abstain · no	0.38	1.00	0.9s	PASS

Eval harness · run #4812

Deploy / rollback surface

Canary on 10% → guardrails on p95 latency, error rate, and grounding score → auto-promote. 90-second rollback SLO.

Five things I didn't expect.

01 **Local-first is a UX decision, not just a security one.**

Operators trust a tool more when they can literally turn their Wi-Fi off and it still works. The privacy story lands emotionally.

02 **Prompt-cache hits are the real latency win.**

The intent + retrieval output is cached per-ticket — second generations are 3× faster. Worth more than model quantization.

03 **Logreg is not a downgrade — it's a feature.**

Inspectable weights mean every routing decision is defensible. 'Why did you send this to the policy lane' has a concrete answer.

04 **Tauri + Rust is the right desktop stack in 2026.**

Bundle size, Apple notarization, and Rust FFI for the ML sidecar made iteration 2-3× faster than the Electron alternative.

05 **The feedback loop only works if rating is one click.**

Anything longer than thumbs-up / thumbs-down gets skipped. All the KB gap data comes from that single-click surface.

● THANKS FOR WATCHING

Questions?

Open source · MIT licensed · runs on any M-series MacBook with Ollama installed.

REPO

github.com/saagpatel/AssistSupport

current repo · v1.2.0 · MIT

DECK + ONE-PAGER

portfolio drop

PDF + slide deck + screenshot set

CONNECT

[in/saagarpatel](https://www.linkedin.com/in/saagarpatel)

DMs open — IT platform + local AI

Built with Tauri 2 · React 19 · TypeScript · Rust · SQLCipher · Ollama · TF-IDF + MiniLM-L-6-v2 · logreg intent classifier